Quiz 3 → ch 7 (upto this lec) next week.

$R(A, B, C, D)$ FD: $\alpha \rightarrow \beta$

$AB^+ = \{A, B, C, D\}$ → AB is a superkey → covers all the attributes

$CD^+ = \{B, C, D\}$

SK ⇒ CK ⇒ More c k

$A^+ = \{A, B, C, D\}$

$A \rightarrow A, B, C, D, E$

$A^+ = \{A, B, C, D, F\}$

$A \rightarrow A$

$A \rightarrow B$

$A \to C$

$A \to D$

$A \to E$

$A \to B$, $D \to E$, $B \to C$

Start with reflexivity.

$AD^+ = \{A, D, B, E, C\}$

Smaller key ? Bigger key

A^+, AD → both superkey

but we take A^+ → ?

Method 3 : $ADE^+ = \{ \ldots \}$

$A \to B$

$C \to D$

static c.

$DA = \{B, D\}$

$PA = \{A, D, F, F$

Dynamically changing.

$\therefore DA \cap PA = \{\}$ → Guarantee no more CK.

Assignment next week → this algo → quiz.

CK output.

$R(A, B, C, D, E)$ F.D $A \rightarrow B$, $D \rightarrow E$

Dependent Attrc ←

$DA = \{B, E\}$

$ABCDE^+ = \{A, B, C, D, E\}$

can determine

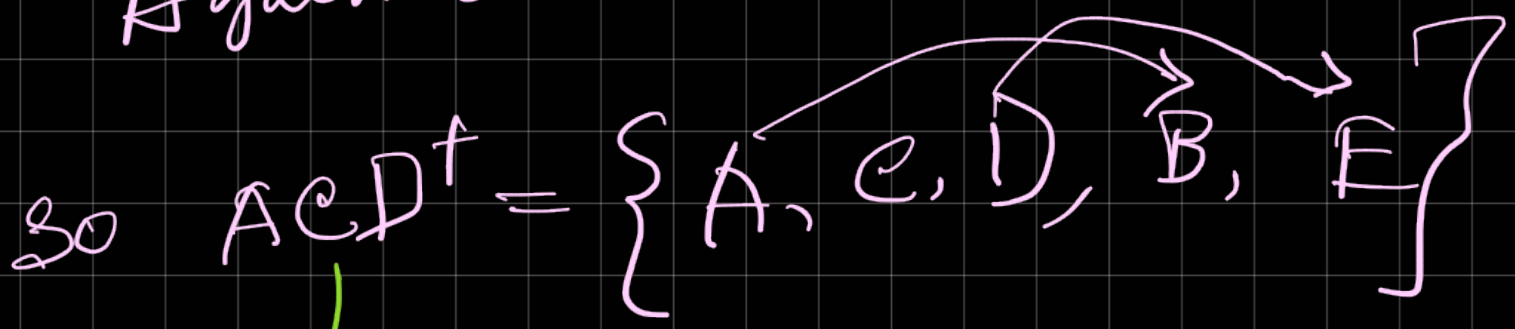
Dept, DeptLoc, ... =

redundant

→ if A can determine B & B is already there, no need for

B, so, ACDE+

Again same for E



RS → Reduced Set.

$2^3 - 2 = 6$ ← superkey

null, full

without reduction $= 2^5 - 2$
$= 32 - 2$
$= 30$!

AcD+ = {all} → only 1 CK

$AC^+ = \{A, C, B\}$ $DA = \{B, E\}$

$CD^+ = \{C, D, E\}$ $PA = \{A, C, D\}$

$AD^+ = \{A, D, B, E\}$ $\alpha = \phi$ (NULL)

$A^+ = \{A\}$

$A^+ = \{A, B\}$ no other CK.

$C^+ = \{C\}$

$D^+ = \{D, E\}$

$R(A, B, C, D)$ FD: $A \rightarrow B$

$B \rightarrow C$

$C \rightarrow A$

$ABCD^+ = \{A, B, C, D\}$

$AD^+ = \{A, B, C, D\}$ → superkey.

$AD^+ = \{A, B, C, D\}$ → SK & CK ✓

$A^+ = \{A, B, C\}$ } no D

$D^+ = \{D\}$

$BA = \{B, C, A\}$

$PA = \{A, D\}$

Dynamic

$\alpha = A$

target → find an α that would determine A.

AD

$C \rightarrow A$

CD (no need to check)

$C^+ =$

$D^+ =$

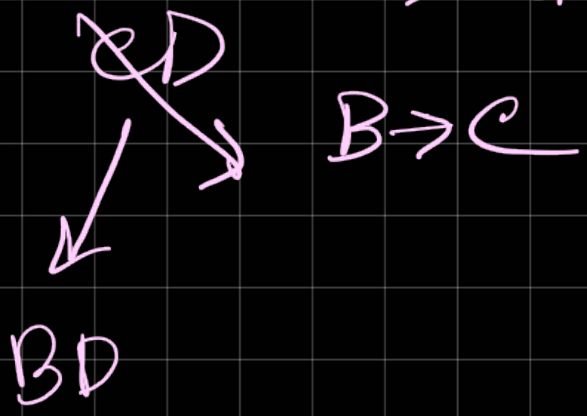
$CK_1 = AD$

$CK_2 = CD$ } process α.

$\alpha = A\,B\,C$ $\quad \alpha \rightarrow A$, $\alpha \rightarrow C$

if any of the process

is repeated, stop.



$$2^2 = 4$$

$$2 \div 2 = 1$$

$$5 \div 2 = 2.5$$

Nice!